AnsyTruss-2D User Manual

MATLAB-based 2D Truss Analysis Tool

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1 Introduction

AnsyTruss-2D is a **MATLAB-based** graphical user interface (GUI) app developed for the static analysis of two-dimensional pin-jointed trusses. The app provides a visual verification stage for checking the defined truss geometry, support conditions, and applied loads before carrying out the analysis using the direct stiffness method. Through editable input tables, users can define the truss geometry, material and sectional properties, support conditions, and applied nodal loads. The app then calculates nodal displacements, member axial forces, and support reactions, and provides visual outputs showing both the verified geometry and the deformed truss shape.

2 Coordinate System and Sign Convention

AnsyTruss-2D uses a global Cartesian coordinate system. The positive X-direction is defined from left to right, and the positive Y-direction is defined from bottom to top. Accordingly, nodal coordinates, applied loads, displacements, and support reactions should be interpreted using this sign convention.

3 Units

AnsyTruss-2D does not assign or convert physical units internally. Therefore, all input values must be entered using a consistent unit system. The calculated displacements, axial forces, and reactions will follow directly from the units used in the input data.

4 Theory of 2D Truss Analysis

AnsyTruss-2D utilises the **Direct Stiffness Method**, also known as the Finite Element Method, to analyse two-dimensional pin-jointed truss structures.

4.1 Truss Element Degrees of Freedom

A truss member is assumed to carry axial force only. Therefore, in the local element coordinate system, each node has one axial displacement degree of freedom along the member axis as shown in Figure 1. For a two-node truss element, the local displacement vector is

$$\mathbf{d}'_e = \begin{bmatrix} u'_i & u'_j \end{bmatrix}^T . \tag{1}$$

Although the member has only axial deformation in its local coordinate system, it is assembled within a two-dimensional global coordinate system. Therefore, each node in the global system is represented by two translational degrees of freedom: the horizontal displacement u and the vertical displacement v. The corresponding global element displacement vector is

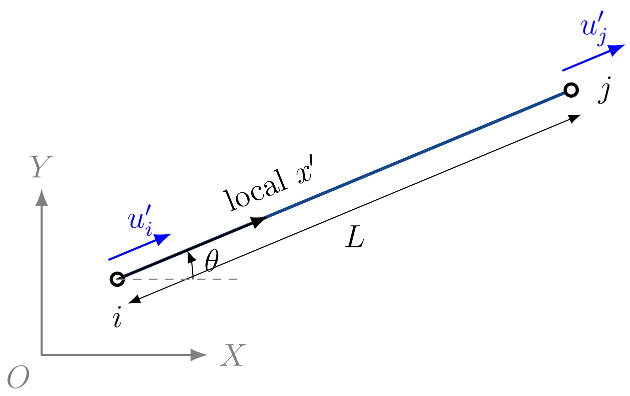


Figure 1: Inclined two-node truss element with axial displacement degrees of freedom in the local coordinate system.

$$\mathbf{d}_e = \begin{bmatrix} u_i & v_i & u_j & v_j \end{bmatrix}^T. \tag{2}$$

4.2 Element Stiffness Matrix

For a truss element with length L, cross-sectional area A, and Young's modulus E, the local axial stiffness matrix is

$$\mathbf{k}' = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \tag{3}$$

For an inclined element connecting node i to node j, the element orientation is described by the direction cosines

$$c = \cos\theta = \frac{x_j - x_i}{L}, \qquad s = \sin\theta = \frac{y_j - y_i}{L}, \tag{4}$$

where

$$L = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}. \tag{5}$$

The local axial deformation is related to the global nodal displacements through the transformation

$$\begin{bmatrix} u_i' \\ u_j' \end{bmatrix} = \begin{bmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{bmatrix} \begin{bmatrix} u_i \\ v_i \\ u_j \\ v_j \end{bmatrix}. \tag{6}$$

Using this transformation, the element stiffness matrix in the global coordinate system is

$$\mathbf{K}^e = \frac{EA}{L}\begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}. \tag{7}$$

4.3 Global System Assembly

The global stiffness matrix for the complete truss is assembled by adding the contributions of all individual elements according to their global degrees of freedom:

$$\mathbf{K} = \sum_{e=1}^{N_e} \mathbf{K}^e. \tag{8}$$

After applying the support boundary conditions, the unknown nodal displacements are obtained from the global equilibrium equation

$$\mathbf{KU} = \mathbf{F}, \tag{9}$$

where $\mathbf{K}$ is the assembled global stiffness matrix, $\mathbf{U}$ is the global nodal displacement vector, and $\mathbf{F}$ is the global nodal force vector.

4.4 Member Axial Force

Once the nodal displacements are known, the axial force in each member is calculated from the displacement difference along the element axis:

$$f = \frac{EA}{L}\begin{bmatrix} -c & -s & c & s \end{bmatrix}\begin{Bmatrix} U_{ix} \\ U_{iy} \\ U_{jx} \\ U_{jy} \end{Bmatrix}. \tag{10}$$

A positive value of f indicates that the member is in tension, while a negative value indicates compression.

5 App Panels

The GUI is organised into three main panels: the **Input Panel**, **Verification Panel**, and the **Results Panel**. These panels are arranged to provide a clear workflow for defining the truss model, verifying the entered geometry and boundary conditions, and reviewing the analysis results.

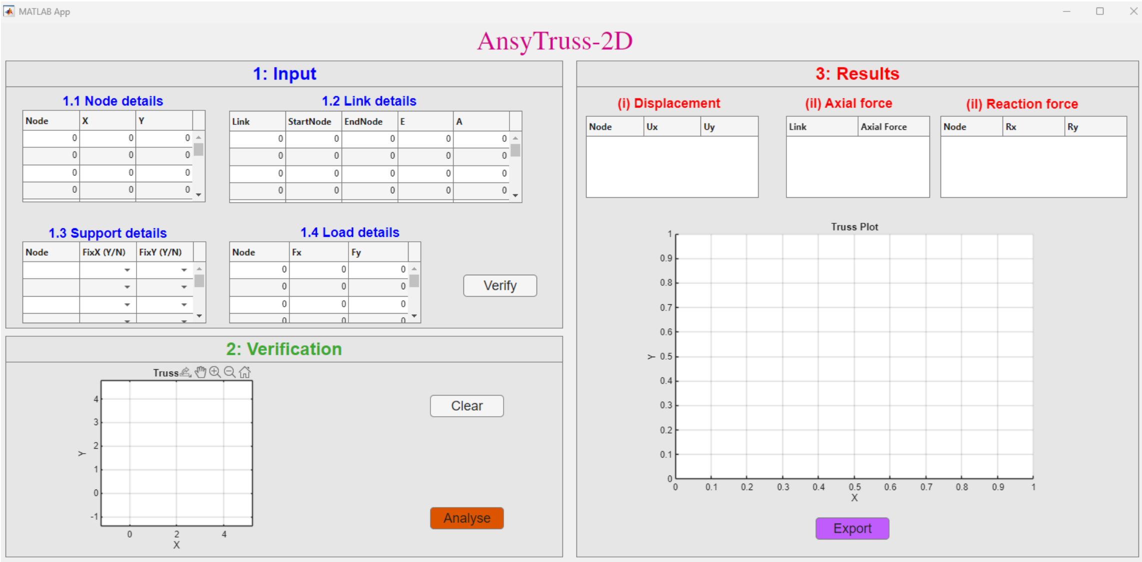


Figure 2: Overview of the AnsyTruss-2D graphical user interface showing the main app panels.

5.1 Input Panel

The Input Panel is used to define the truss model. It allows the user to enter the required model data, including node coordinates, link connectivity, support conditions, and applied loads. The four main components are:

- Node Data
- Link Data
- Support Data
- Load Data

Users can type the required values directly into the tables. Blank rows may be left unused.

5.2 Verification Panel

The Verification Panel is used to check the truss model before analysis. After clicking the **Verify** button, the app plots the initial truss geometry and displays the entered model visually. The verification plot displays:

- Node numbers
- Link numbers
- Pinned and roller support symbols
- Applied loads with appropriate directional arrows

This step helps confirm that the geometry, connectivity, support conditions, and applied loads have been entered correctly before running the analysis. The **Clear** button is used to clear the entered input data from the input tables.

5.3 Results Panel

The Results Panel displays the outputs of the truss analysis. After clicking the **Analyse** button, the app shows:

- Nodal displacement table
- Member axial force table
- Reaction force table
- Deformed truss shape overlaid on the initial undeformed geometry, with axial force contours displayed along the members

The **Export** button is provided to generate and save the analysis report in PDF format.

6 Preloaded Example and Input Data Format

A simple three-node triangular truss is provided as a preloaded demonstration example in the software (Figure 3). This example is useful for quickly understanding the data format and testing the **Verify**, **Analyse**, and **Export PDF Report** functions.

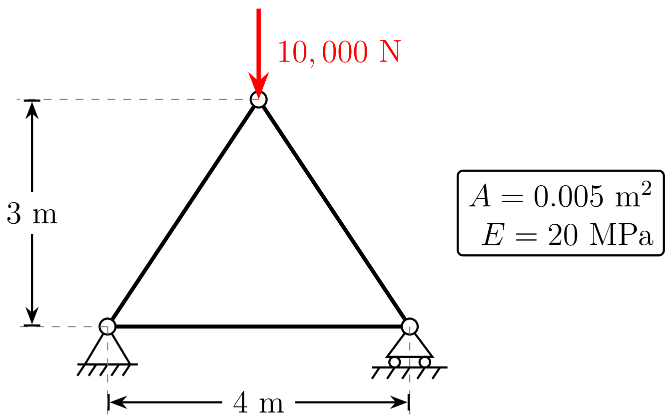


Figure 3: A 2D Pin-Jointed triangular truss with cross-sectional area $A = 0.005$ m^2 and Young's modulus $E = 20$ MPa.

6.1 Fixing the Coordinates

To mathematically model the truss, we must establish a global coordinate system. Let us place the origin $(0, 0)$ at the bottom-left pin support. Because the truss has a 4 m span and is symmetric, the top node is located at the horizontal midpoint ($X = 2$ m) and a height of $Y = 3$ m.

6.2 Assigning Node and Link Numbers

Before entering data, you must assign a unique **Node number** to every joint and a unique **Link number** to every member. For this example, we assign:

- **Node 1:** Bottom-left support

- **Node 2:** Bottom-right support
- **Node 3:** Top apex

These assignments dictate how the following input tables are filled out.

6.3 Node Data Format

The Node Data table defines the position of each node based on your coordinate system.

Node	X	Y
1	0.0	0.0
2	4.0	0.0
3	2.0	3.0

Columns: `Node` (Node number), `X` (Horizontal coordinate), `Y` (Vertical coordinate).

6.4 Link Data Format

The Link Data table defines the truss members and their connectivity.

Link	StartNode	EndNode	E	A
1	1	2	2.0×10^7	0.005
2	1	3	2.0×10^7	0.005
3	2	3	2.0×10^7	0.005

Columns: `Link` (Element number), `StartNode` and `EndNode` (The nodes it connects), `E` (Young's modulus), `A` (Cross-sectional area).

6.5 Support Data Format

The Support Data table defines restrained and free degrees of freedom.

Node	FixX	FixY
1	Y	Y
2	N	Y

Use `Y` to signify a restrained (fixed) degree of freedom, and `N` for a free degree of freedom. For example, Node 1 is a pin (`Y, Y`) and Node 2 is a horizontal roller (`N, Y`).

6.6 Load Data Format

The Load Data table defines nodal loads.

Node	Fx	Fy
3	0.0	-10000.0

Sign convention: Positive `Fx` acts to the right, negative `Fx` acts to the left. Positive `Fy` acts upward, negative `Fy` acts downward.

7 Using the App

Step 1: Enter Input Data

Enter your mapped node data, link data, support data, and load data into the corresponding tables in the Input Panel (see Figure 4). **Use consistent units throughout the model**. For example, if length is entered in metres and force in Newtons, then Young's modulus and area must also be entered in compatible units (Pascals and m^2).

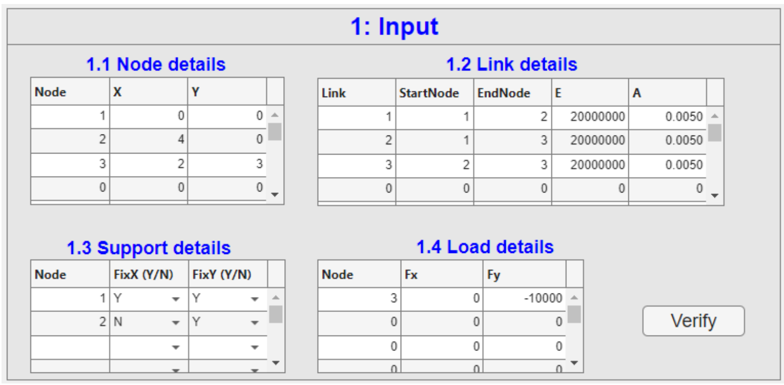


Figure 4: Input panel populated with the data from the preloaded example.

Step 2: Verify the Truss

Click the **Verify** button. The app checks the input data and plots the verification geometry (see Figure 5). The verification stage checks for empty tables, duplicate nodes, invalid start/end nodes, zero-length links, and invalid support/load locations. If the displayed geometry does not appear correct, return to the input tables and edit the data.

Step 3: Analyse the Truss

Click the **Analyse** button. The app solves the truss and displays the results in the Results Panel (see Figure 6). The final plot shows the deformed truss shape with an axial force contour. Positive axial force indicates tension, while negative axial force indicates compression.

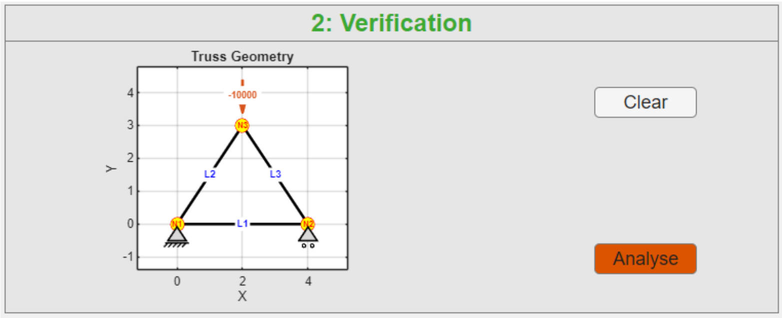
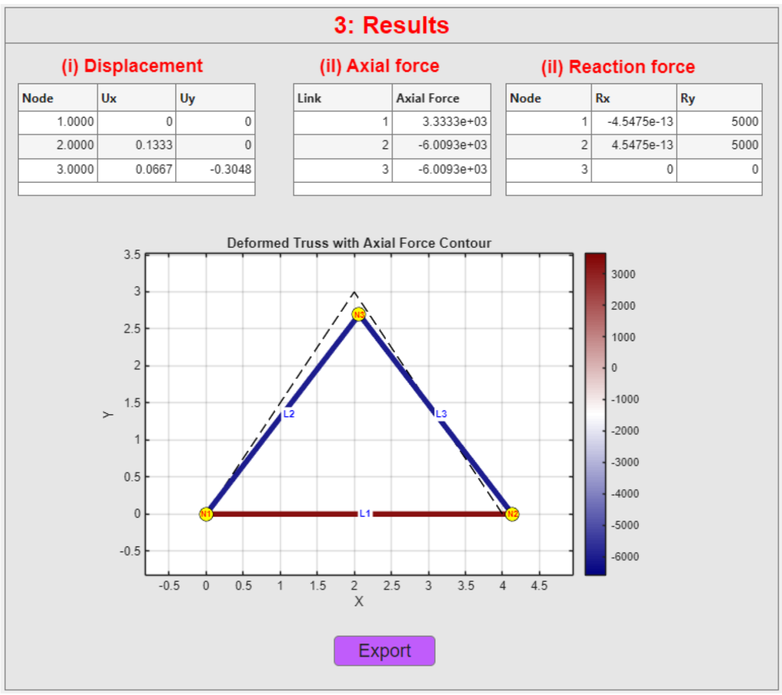


Figure 5: Verification panel showing Node numbers, Link numbers, supports, and applied loads.



Node	Ux	Uy
1.0000	0	0
2.0000	0.1333	0
3.0000	0.0667	-0.3048

Link	Axial Force
1	3.3333e+03
2	-6.0093e+03
3	-6.0093e+03

Node	Rx	Ry
1	-4.5475e-13	5000
2	4.5475e-13	5000
3	0	0

Figure 6: Results panel displaying tabular outputs alongside the deformed and undeformed configurations.

Step 4: Export PDF Report

Click the **Export PDF Report** button to generate an A4 PDF report. Wait for the pop-up message saying **"PDF report exported successfully."** before opening the file.

8 Output Results

8.1 Displacement

The nodal displacement table contains **Ux** (horizontal displacement) and **Uy** (vertical displacement).

8.2 Axial Force

The axial force table outputs the internal force per link. Positive force indicates tension, and negative force indicates compression.

8.3 Reaction Force

The reaction table contains **Rx** (horizontal reaction) and **Ry** (vertical reaction).

9 Exporting the Report

The **Export** button creates a two-page PDF report summarising the truss model and analysis results.

- Page 1 contains the verification plot and input tables, including node data, link data, support data, and load data.
- Page 2 contains the deformed truss shape with axial force contours, displacement results, axial force results, and support reaction results.

The exported report includes:

- Verified truss geometry
- Node, link, support, and load input tables
- Deformed truss shape overlaid on the undeformed geometry
- Axial force contour plot
- Nodal displacement table
- Member axial force table
- Support reaction table

10 Assumptions and Limitations

AnsyTruss-2D is developed for the static analysis of two-dimensional, pin-jointed truss structures. The following assumptions are adopted:

- The structure is a 2D pin-jointed truss and is in static equilibrium.
- Members carry axial force only; bending moments, shear forces, and torsional effects are not considered.
- Material behaviour is linear elastic and deformations are assumed to be small.
- Loads are applied only at nodes and connections are ideal frictionless pins.
- Geometric nonlinearity, material nonlinearity, buckling, and dynamic effects are not considered.

11 Troubleshooting

Problem	**Possible Solution**
Verification fails	Check that node numbers are unique and that all link start/end nodes exist.
Analysis does not run	Check supports, loads, and ensure the truss is stable.
No load arrow appears	Check that the load node exists and that Fx or Fy is not zero.
Unexpected displacement values	Check units for E, A, length, and force.
PDF does not open correctly	Wait for the export-success pop-up before opening the file.

12 Summary

AnsyTruss-2D provides a simple and structured workflow for defining, verifying, analysing, visualising, and reporting two-dimensional truss structures. As a MATLAB-based GUI tool, it supports static analysis of linear elastic pin-jointed trusses using the direct stiffness method. It is developed at Auckland University of Technology (AUT) as an educational and engineering calculation tool for truss analysis.